

Domain for x : Set of all users
 Domain for y : Set of all websites
 Domain for z : Set of all urls

$\exists x \forall y A(x, y, \text{edu})$

d) There are exactly two systems that monitor every remote server

$M(x, y)$: system x monitors remote server y

Domain for x : Set of all systems

Domain for y : Set of all remote servers

$\exists x \exists y \forall z ((x \neq y) \wedge M(x, z) \wedge M(y, z) \wedge (M(w, z) \Rightarrow (w = x \vee w = y)))$

19) Express each of these statements using mathematical and logical operators, predicates and quantifiers, where the domain consists of all integers.

a) The sum of two negative integers is negative: $\forall x \forall y ((x < 0) \wedge (y < 0) \Rightarrow (x + y < 0))$

b) The difference of two positive integers is not necessarily positive: $\neg (\forall x \forall y ((x > 0) \wedge (y > 0) \Rightarrow (x - y > 0)))$

c) The sum of the squares of two integers is greater than or equal to the square of their sum: $\forall x \forall y ((x^2 + y^2) \geq (x + y)^2)$

d) The absolute value of the product of two integers is the product of their absolute values: $\forall x \forall y (|x \cdot y| = |x| \cdot |y|)$

20) Express each of these statements using predicates, quantifiers, logical connectives and mathematical operators where the domain consists of all integers.

a) The product of two negative integers is positive: $\forall x \forall y ((x < 0) \wedge (y < 0) \Rightarrow (xy > 0))$

b) The average of two positive integers is positive: $\forall x \forall y ((x > 0) \wedge (y > 0) \Rightarrow (x + y) / 2 > 0)$

c) The difference of two negative integers is not necessarily negative: $\neg \forall x \forall y ((x < 0) \wedge (y < 0) \Rightarrow (x - y < 0))$